

282 tickers. 10 years. 83,422 simulated trades.

Everything we tested for the Alpaca hackathon, what survived, and the rules the agent runs on. Every number below is reproducible from the repository.

282 TICKERS TESTED	83,422 TRADES SIMULATED	40 PASSED THE SCREEN	28 GOLDEN RULES
-----------------------	----------------------------	-------------------------	--------------------

How the screen narrowed

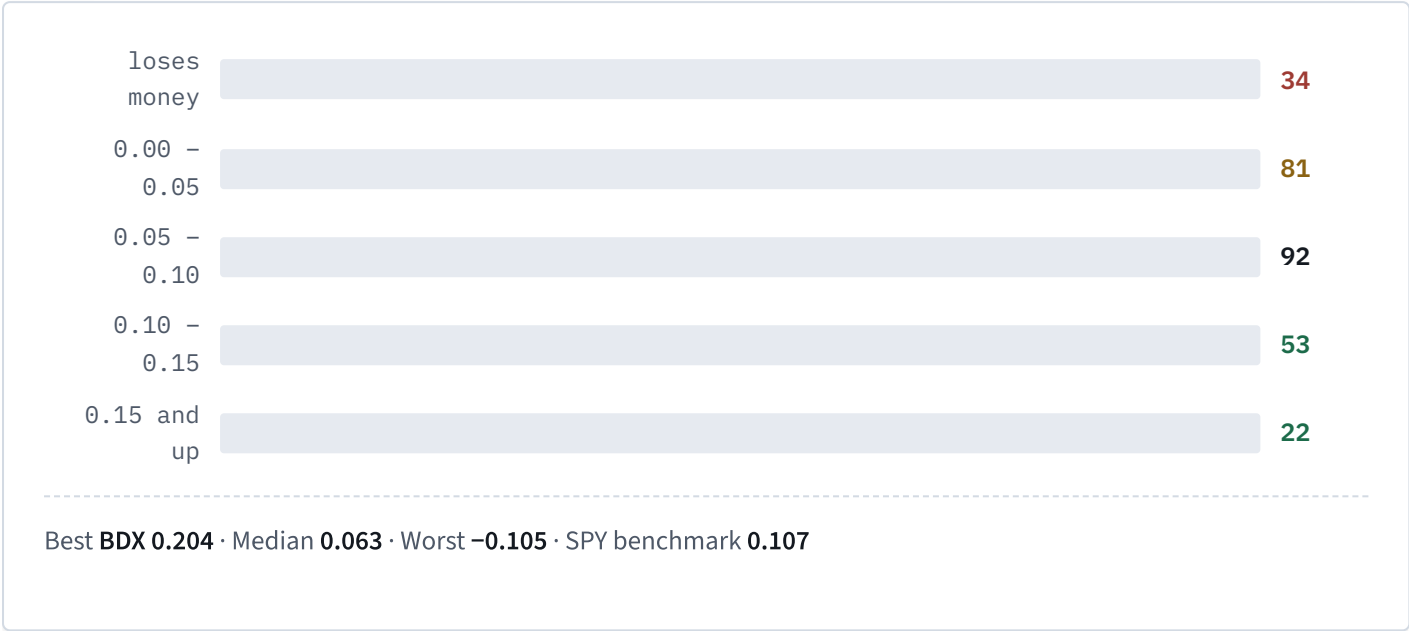
Each ticker ran through the same ten-year backtest of our iron-condor strategy — about 295 simulated trades per name. A ticker had to beat SPY on *both* profitability and drawdown to survive.

287 SUBMITTED Full list across every sector.	217 LOST TO SPY Less profit per unit of risk than simply trading the index.	24 BACKUP TIER Beat SPY, but drawdown too deep to hold a core slot.	40 PASSED Beat SPY on profit <i>and</i> risk. These are what we trade.
--	---	---	--

5 more were skipped — ARM, GEV, RDDT, TEM and CRCL are too newly listed to have enough history to test.

Where the whole field landed

Expectancy is profit per unit of risk. Every tested ticker, grouped. SPY sits at 0.107 — better than 77% of the field, which is why it is the bar.



The 40 that passed

Ranked by expectancy. The top five fill our trading slots first. Drawdown is the worst peak-to-trough run over ten years, in units of spread width.

#	TICKER	EXP.	WIN%	DRAW
1	BDX	0.204	74.2%	-2.70
2	PGR	0.201	75.6%	-3.34
3	ET	0.188	71.9%	-3.43
4	MA	0.186	71.2%	-3.02
5	CB	0.186	71.2%	-2.33
6	RTX	0.183	73.9%	-4.48
7	TDG	0.182	73.6%	-4.32
8	KO	0.180	71.9%	-4.48
9	GILD	0.179	71.3%	-3.28
10	CINF	0.179	73.0%	-3.34
11	OXY	0.172	71.9%	-3.45
12	ARCC	0.167	71.7%	-3.17
13	SBUX	0.166	72.3%	-3.55
14	AVGO	0.163	71.3%	-2.78
15	V	0.163	71.2%	-3.28
16	ALL	0.160	70.9%	-3.24
17	BBY	0.154	70.9%	-4.32
18	VICI	0.150	70.9%	-3.82
19	CSCO	0.149	70.7%	-3.43
20	CVS	0.147	71.2%	-2.95
21	TJX	0.147	70.6%	-3.38
22	PEP	0.146	69.0%	-3.50
23	HON	0.144	70.6%	-3.45
24	AOS	0.140	71.2%	-3.32
25	LRCX	0.138	72.3%	-2.94
26	AEP	0.134	70.7%	-3.15
27	MDT	0.134	71.2%	-3.45
28	COF	0.132	71.2%	-4.43
29	ELV	0.132	70.6%	-4.42
30	CEG	0.132	68.6%	-2.52
31	ABT	0.131	69.2%	-4.48
32	LVS	0.130	70.2%	-4.24
33	CVX	0.126	68.9%	-4.11
34	GM	0.120	69.6%	-3.71
35	STT	0.119	70.2%	-4.06
36	JNJ	0.117	67.6%	-4.03
37	HOG	0.115	69.6%	-4.02
38	KLAC	0.114	69.3%	-4.13
39	XOM	0.114	67.6%	-4.36
40	ICLR	0.111	69.3%	-3.78
—	SPY	0.107	67.3%	-3.39

The pattern: boring wins. Insurance, medical devices, pipelines, staples, defense, utilities. Every high-growth name lost to the index — PLTR, COIN, NET, SNOW, CRWD, SMCI, RBLX, HOOD, NVDA, TSLA, META, AAPL. They pay more premium, but they gap harder, and the tail costs more than the premium is worth.

Backup tier — 24 names

Beat SPY on profit but with drawdown deeper than -4.5. Used only when the core 40 have nothing qualifying.

NAMES
BMJ · ACGL · ADI · UNP · MKC · HIG · CI · WMB · DE · BALL · KMI · HLT · APH · FE · GIS · EQT · CMI · LYB · DKS · BSX · and 4 others

The strategy

We sell volatility, not direction. Four directional strategies were tested and all four failed, so the agent takes no view on which way the market moves.

PARAMETER	SETTING	WHY
Structure	Iron condor — sell both sides	Defined risk; no directional edge proven, so no side taken
Expiry	14 days	Past the gamma danger zone, short enough for decay to accrue
Short strikes	0.20 delta	~2% out of the money — best tested configuration
Minimum credit	1.15 × delta × width	Below this the structure loses money — measured, not guessed
Exit	day 7 at 50% of credit	Halves exposure; lifts win rate from 56% to 67%
Per position	2% of equity	\$2,000 max loss on \$100,000
Portfolio ceiling	10% of equity	Worst possible week ends at \$90,000 — arithmetic, not forecast
Concurrent slots	5	Highest-ranked names fill first

The 12 risk gates

Every candidate passes all twelve or is refused and logged with the reason. Pure code — no model output can override them.

GATE	REJECTS
quote_sanity	Crossed or stale market data
tradeable	Halts and corporate actions
defined_risk	Unbounded exposure
dte	ODTE and anything too far out
earnings	Gap risk over a known event
liquidity	Strikes you cannot exit

GATE	REJECTS
min_credit	Trades fees would consume
spread_quality	Wide markets that tax entry and exit
credit_fraction	Premium too thin for the risk
reward_risk	Poor payoff on directional trades
position_size	Oversized positions
portfolio_risk	Concentration across the book

The 28 golden rules

OPTIONS — FROM THE RESEARCH CORPUS

R1	Never sell a put unless you'd be content to own the stock at that strike
R2	Own the collateral before selling against it
R3	Select short strikes inside the 15–35 delta band
R4	Size and timing must respect the decay curve
R5	ODTE is out of scope — hard constraint

PROCESS — RISK AND VALIDATION

S0	Trade only where expectancy is positive — never on win rate alone
S1	Minimum reward-to-risk floor, defined before entry
S2	Size from defined risk, at a constant fraction
S3	Backtest → simulator → live at minimum size → scale
S4	Log every evaluation, including the ones we refuse
S5	Kill-switch thresholds come from the backtest, not intuition
S6	Not trading is a position
S7	Screen the underlying before analysing it

EXECUTION — DISCOVERED BY BUILDING

E1	Enter where the book is deep — 09:45–10:30 and 14:30–15:15 ET only
E2	Most active ≠ best to enter. The busiest hour is for exits
E3	Payoff gates must be structure-aware — credit and debit need different tests
E4	Triggers nominate; gates decide. No human or model places an order
E5	Exits are placed at entry, as resting orders
E6	Branches are decided before they arrive
E7	Raise risk only where the floor stays arithmetic
E8	Sell premium when it's rich — mornings
E9	Time-of-day is hygiene, never signal
E10	Classify before encoding — structural, empirical, or operational
E11	Direction-neutral until a directional edge is proven
E12	Idle capital goes to the strategy that needs no edge
E13	Validate the data before calibrating against it
E14	Test the payoff, not the instrument

What we tested and threw away

Four strategies measured against a proper baseline. None survived. Recording the failures is what makes the surviving rules credible.

STRATEGY	SAMPLE	RESULT
VWAP regime filter → direction	2,679 days	No separation; the sign ran opposite to the hypothesis
EMA / MACD / volume signal	32,148 bars	Original threshold showed no edge; tightened version failed correction
20-day high breakout	1,527 signals	+0.17% vs base rate — not significant
Breakout + clear uptrend	1,103 signals	−0.02% — the extra filter made it <i>worse</i>

Last completed week

19–28 AUGUST · TOP 5 NAMES · \$100,000 ACCOUNT

+\$4,459

+4.46%. Five positions, five winners. RTX and MA both breached their short strikes and still finished green — the credit covered the loss.

Read this before celebrating. That was one week, and a good one. The ten-year data says roughly **one week in three loses money**. A typical week is +1–2%, not +4.5%. We are building a defensible result, not a lottery ticket — and P&L is one of five judging criteria.

Method — ten years of daily bars per ticker (2016–2026), ~295 simulated trades each, 83,422 in total. Iron condor at 0.20 delta, 14 days to expiry, credit assumed at the gate minimum, exit on day 7 at 50% of credit.

Expectancy = $(1 + \text{average win} \div \text{average loss}) \times \text{win rate} - 1$. Drawdown is the worst peak-to-trough run, in units of spread width.

Screening also excludes any name with earnings inside the contract's life, or with fewer than five strikes carrying real open interest. 198 automated tests passing.